ILLUSTRATIVE EXAMPLES

5.2 EQUATIONS REDUCIBLE TO QUADRATIC FORM

Example Statements (as per textbook)

Example 1. Find the roots of the equation

$$\frac{1}{x+1}+\frac{2}{x+2}=\frac{4}{x+4}.$$

Example 2. Solve:

$$\frac{x-1}{x-2}-\frac{x-2}{x-3}-\frac{x-5}{x-6}-\frac{x-6}{x-7}.$$

Example 3. Solve:

$$\frac{a}{ax+1}+\frac{b}{bx+1}=a+b, \quad ab\neq 0,\ a+b\neq 0.$$

Example 4. Solve:

$$2^{2x+8}+1=32\cdot 2^x.$$

Example 5. Solve:

$$(x^2-5x)^2-30(x^2-5x)-216=0.$$

Example 6. Solve:

$$(x+1)(x+2)(x+3)(x+4)=120.$$

Example 7. Solve:

$$x^{2/3}+x^{1/3}=2.$$

Example 8. Solve the equation

$$8\sqrt{\frac{x}{x+3}}-\sqrt{\frac{x+3}{x}}=2.$$

Example 9. Solve the equation

$$\sqrt{a+x}+\sqrt{a-x}=a, \quad x\neq 0.$$

Example 10. Solve

$$\sqrt{x}=x-2.$$

Example 11. Solve the equation

$$\sqrt{x^2+3x+32}+\sqrt{x^2+3x+5}=9.$$

Example 12. Solve the equation

$$\sqrt{x+5}+\sqrt{x+21}=\sqrt{6x+40}.$$

Example 13. Solve the equation

$$\left(x+\frac{1}{x}\right)^2=4+\frac{3}{2}\left(x-\frac{1}{x}\right).$$

Example 14. Solve

$$4x^4-16x^3+7x^2+16x+4=0.$$

Example 15. Solve:

$$(x+2)(x+3)(x+8)(x+12)=4x^2.$$

Final Answers

Example 1. $2+2\sqrt{3},\ 2-2\sqrt{3}$

Example 2. $x=4\frac{1}{2}$

Example 3. $0,\ -\dfrac{a^2+b^2}{ab(a+b)}$

Example 4. $x=-4$

Example 5. $9,\ -4,\ 3,\ 2$

Example 6. $1,\ -6,\ \dfrac{-5\pm i\sqrt{39}}{2}$

Example 7. $1,\ -8$

Example 8. $x=1$

Example 9. $a,\ -a$

Example 10. $x=4$

Example 11. $-4,\ 1$

Example 12. $x=4$

Example 13. $-\dfrac{1}{2},\ -1,\ 1,\ 2$

Example 14. $2,\ -\dfrac{1}{2},\ \dfrac{5\pm\sqrt{41}}{4}$

Example 15. $-4,\ -6,\ \dfrac{-15\pm\sqrt{129}}{2}$